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# Extending Functional Worst-Risk Minimization to Non-Linear Models

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*Abstract*

Imagine a functional structural system where your target variable $Y$ is determined by your predictor variable $X_1$ and subsequently generates or causes the variable $X_2$ ($X_1 \to Y \to X_2$). This functional system is observed in 2 environments, the observational and interventional, subject to external shifts.

Standard predictive models often achieve minimal error by accounting for all variables correlated with $Y$, including spurious ones, like $X_2$ that is correlated with $Y$ but is not causing it. However, the moment you move to a shifted environment, for example move to a different market regime, your prediction is not working anymore. The only correlation that lives in all environments is the invariant causal relationship (In this case $X_1 \to Y$)

The topic of Causal Inference and the methods of finding such relationships vary from different problem and constraint settings. Kennerberg and Wit [2025] work on "Functional worst risk minimization" was concluded with the worst-risk decomposition and the estimation of optimal future worst risk minimizer in the scope on Functional Linear relationships. The main goal of this thesis is to extend the functional worst-risk minimization framework to non-linear systems using a Functional Additive Regression (FAR) model from Fan et al. [2015].

As we observe in the course of work, the transition from the Linear model to the Non-Linear introduced a hurdle of the distributive failure, the exact worst-risk decomposition is no more possible.

In a non-linear system, an unseen interventional shift could theoretically interact with the steep gradients of the non-linear function to inflate the prediction error toward infinity. Our main theoretical result manages this issue by establishing a tight mathematical upper bound, proving that the error is strictly contained.

The empirical part of the thesis was focused on finding the best algorithm that estimates underlying relationships between variables with observational and interventional environments at disposal. Due to the mentioned inability of the direct risk decomposition, we needed to settle the Empirical formulation for the worst risk, along the hyper-parameter $\gamma \in [0, 1]$ that manages the trade-off between the observed environments.

To solve this complicated non-convex optimization problem, the proposed algorithm was broken into three decoupled phases. It starts with an alternating block coordinate descent and fast polynomial profiling to safely navigate the data and lock in the projection kernels. With the directions fixed, it uses a cross-validated Group Lasso to drop false variables entirely, leaving a clean path to fit an unpenalized natural spline that reveals the true non-linear effect.

As a result, the estimation procedure recovers the exact structural geometry and maintains robust predictive accuracy even under severe interventional shifts.

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# 1 Introduction

Statistical learning methods typically assume that training and test data originate from the same distribution, which enables effective risk minimization. However, real-world applications frequently involve distributional shifts, leading to poor model generalization [Peters et al., 2015, Arjovsky et al., 2020, Rothenhäusler et al., 2020, Kania and Wit, 2025]. To address this challenge in functional data, Kennerberg and Wit [2025] introduced a functional worst-risk minimization approach for functional structural equation models (SEMs). This approach ensures robust predictive performance across shifted environments, however being limited to the linear relations between variabes. Therefore, the non-linear extension of the work was needed, and the core mathematical challenge of this thesis lies in transitioning this framework from linear to non-linear functional risk minimization. Furthermore we try to follow the structure of the preceding work as much as possible, to make the extension precise and rigorous.

Looking in details at the original framework proposed by Kennerberg and Wit, the functional causal operator is assumed to be strictly linear. The prediction of the target variable $Y$ is given by a linear integral operator:

$$\hat{Y}_t = \int_{T_1}^{t} \beta(t,\tau)X(\tau)d\tau$$

Because this operator is linear, the worst-case expected risk over a bounded set of future shifts $A'$ can be decomposed as follows:

$$\sup_{A' \in C_A^\gamma(A)} R_{A'}(\beta) = \frac{1}{2}R_+(\beta) + \left(\gamma - \frac{1}{2}\right)R_\Delta(\beta) \quad \textit{for every } \beta \in (L^2([T_1,T_2]))^p$$

This exact mathematical decomposition along with the identifiability of the linear operator allows for a closed-form analytical solution where the optimal projection kernel is found globally via a single matrix inversion.

$$\begin{bmatrix} C_1^\gamma \\ C_2^\gamma \end{bmatrix} = [\gamma G^A + (1-\gamma)G^O]^{-1}[\gamma Z^A + (1-\gamma)Z^O]$$

Once we calculate these matrices, we plug them into our basis functions to get the final functional coefficients:

$$\beta_{x_j y}^\gamma(t,\tau) = \phi^t(t)C_j^\gamma\phi(\tau)$$

However, assuming the effect of one variable on another is a straight line completely fails to capture complex dynamics, such as diminishing returns or threshold effects. This shows the demand of the functional extension of the framework that would allow us to model complex real-life like behaviors.

To capture these non-proportional effects this thesis uses the Functional Additive Regression (FAR) single-index operator introduced by Fan et al. [2015], which projects functional covariates into a scalar index before applying a non-linear link function $g_j$:

$$\hat{Y}_t = \sum_{j=1}^{p} g_j\left(\int_{T_1}^{T_2} \beta_j(t,\tau)X_\tau(j)d\tau\right)$$

This operator perfectly overcomes the curse of dimensionality introduced by the more theoretically full but complex operator definitions, but on the other hand it transforms the worst-risk minimization into a highly non-convex optimization problem. Because the distributional shift $A$ is now mathematically nested inside the unknown non-linear function $g$, the distributive property is not working anymore as it was before:

$$g(\langle\beta, X + A\rangle) \neq g(\langle\beta, X\rangle) + g(\langle\beta, A\rangle)$$

Consequently, the interventional risk cannot be mathematically isolated, meaning a closed-form exact risk decomposition is impossible.

To overcome this non-linear barrier, the thesis tries to find an alternative to the functional worst-risk minimization for the FAR operator. Because an exact algebraic decomposition is intractable, we instead establish a tight upper bound for the worst-case prediction risk of any candidate non-linear predictor, thus proving that the error is not exploding to infinity. We then translate this theoretical bound into an empirical joint optimization problem and implement an iterative algorithm to safely traverse the non-convex loss landscape. This approach tries to effectively overcome the local optimization issue executing non-linear causal discovery as precise as possible.

# 2 Functional structural equation models

In this section we will define a functional structural system that may be subject to distributional shifts, which are referred to as *environments*. Both the target function $Y$ and the other functions $X$ are defined on some compact interval $[T_1, T_2]$, with left and right endpoints $T_1$ and $T_2$ respectively. We will denote $T = T_2 - T_1$.

## 2.1 Linear Functional Structural Equation Models

The "Functional Worst Risk Minimization" paper by Kennerberg and Wit [2025] defines the linear version of the Functional SEMs that is then used in the further proofs and sections. Unlike previous models defined in "score space" [Müller and Yao, 2008], this system is built on modelling the relationship directly with the linear operator $\mathscr{T}$.

Let $\mathscr{T} : D(\mathscr{T}) \to L^2([T_1, T_2])^{p+1}$, where $D(\mathscr{T}) \subseteq L^2([T_1, T_2])^{p+1}$, be any operator such that

- $\overline{\mathsf{Range}(I-\mathscr{T})} = L^2([T_1, T_2])^{p+1}$; note that this is trivially true if $\mathsf{Range}(I-\mathscr{T})$ is onto $L^2([T_1, T_2])^{p+1}$,
- $\mathscr{R} := \mathsf{Range}(I-\mathscr{T})$ is a Lusin space; this is again true if $\mathsf{Range}(I-\mathscr{T})$ is onto $L^2([T_1, T_2])^{p+1}$, but also if $\mathscr{R}$ is a polish space.
- $\mathscr{S} := (I-\mathscr{T})^{-1}$, is bounded and linear — as well as injective by definition — on $\mathsf{Range}(I-\mathscr{T})$.

The general definition of the functional linear systems is:

$$(X, Y) = \mathscr{T}(X, Y) + \epsilon \tag{1}$$

And by taking the solution operator $\mathscr{S}$ as:

$$\mathscr{S} := (I-\mathscr{T})^{-1} \tag{2}$$

We will have:

$$(Y, X) = \mathscr{S}(\epsilon) \tag{3}$$

And by representing these definitions for the specific observational and shifted environments:

$$(Y^O, X^O) = \mathscr{T}(X^O, Y^O) + \epsilon_O \tag{4}$$

$$(Y^O, X^O) = \mathscr{S}(\epsilon_O) \tag{5}$$

$$(Y^A, X^A) = \mathscr{T}(X^A, Y^A) + A + \epsilon_A \tag{6}$$

$$(Y^A, X^A) = \mathscr{S}(A + \epsilon_A) \tag{7}$$

## 2.2 Non-Linear Functional Structural Equation Models

In this project, we aim to extend the linear functional structural equation model to accommodate non-linear functional relationships.

Such systems must be built on a non-linear operator instead of the linear $\mathscr{T}$. To capture complex, non-proportional effects between functional processes and variables, we transition from strictly linear integral operators to the Functional Additive Regression (FAR) single-index framework, preserving the ability of estimation.

## 2.3 General Definition of Non-Linear Systems

In the linear setting, the structural relationship is defined by a linear operator $\mathscr{T}$. To introduce non-linearity, we replace $\mathscr{T}$ with the non-linear structural operator $\mathscr{T}_{NL}$.

The $\mathscr{T}_{NL}$ operator relies on a single-index architecture. For a system with $p$ functional covariates $X(1), \ldots, X(p)$, the structural effect on the target process $Y_t$ at time $t$ is modeled by first projecting the functional covariate into a scalar index via an unknown bivariate kernel $\beta_j$, and subsequently applying an unknown, non-parametric smooth link function $g_j$ [James and Silverman, 2004]:

$$[\mathscr{T}_{NL}(X)]_t = \sum_{j=1}^{p} g_j \left( \int_{T_1}^{T_2} \beta_j(t, \tau) X_\tau(j) d\tau \right) \tag{8}$$

Unlike the linear operator $\mathscr{T}$, the non-linear operator $\mathscr{T}_{NL}$ is strictly non-invertible. The integration acts as a dimensionality reduction (compressing an infinite-dimensional curve into a scalar), which, combined with the flexible non-monotonicity of $g_j$, creates a many-to-one mapping.

Consequently, a global inverse operator $\mathscr{S}_{NL} := (I-\mathscr{T}_{NL})^{-1}$ does not exist. For the non-linear SEM to be mathematically well-defined and uniquely solvable, we must strictly assume that the structural equations form a Directed Acyclic Graph (DAG). Under the DAG assumption, endogenous variables can be uniquely determined from the exogenous noise and shifts through sequential substitution, bypassing the need for a global inverse map.

While the DAG assumption ensures unique system solvability at the population level, the single-index architecture inherently introduces scale and location ambiguities during empirical estimation. The specific identifiability constraints required to accurately recover $\beta_j$ and $g_j$ are detailed in Section 4.1.

### 2.4 Functional System Observed in Two Environments

We observe the functional system under different distributional environments. Let $\epsilon \in L^2([T_1, T_2])^{p+1}$ be the centralized system noise.

In the observational environment, the data is generated strictly according to the underlying structural relationships and the system noise:

$$(Y^O, X^O) = \mathscr{T}_{NL}(Y^O, X^O) + \epsilon_O \tag{9}$$

In the shifted environment, the system is subjected to an external distributional shift $A \in L^2([T_1, T_2])^{p+1}$. We model the shift $A$ as an additive intervention on the exogenous variables prior to the non-linear transformations:

$$(Y^A, X^A) = \mathscr{T}_{NL}(Y^A, X^A + A) + \epsilon_A \tag{10}$$

### 2.5 The Distributive Failure

In the linear framework proposed by Kennerberg and Wit, the risk under an intervention decomposes algebraically. Because the projection operator is linear, the shift distributes perfectly: $\mathscr{T}(X + A) = \mathscr{T}(X) + \mathscr{T}(A)$.

Under the $\mathscr{T}_{NL}$ operator, the shift is nested inside the non-linear link function $g$. Let the inner product projection be denoted as $\langle \beta, X \rangle$. Because $g$ is non-linear, the distributive property is fundamentally violated:

$$g(\langle \beta, X + A \rangle) \neq g(\langle \beta, X \rangle) + g(\langle \beta, A \rangle) \tag{11}$$

Crucially, the way the interventional shift $A$ inflates the system risk depends entirely on the local shape and gradient of the non-linear function $g$ at the exact coordinates of the projection. This mathematically entangles the shift with the baseline variance, preventing an exact affine risk decomposition. We must therefore establish a worst-case expected risk upper bound instead.

## 3 Functional Worst Risk Decomposition

Because an exact algebraic decomposition is intractable under $\mathscr{T}_{NL}$, we establish a tight upper bound for the worst-case prediction risk of any candidate non-linear predictor $\mathscr{M}$.

We evaluate robustness against unseen shifts bounded by a specific budget $\gamma > 0$ relative to our observed shift $A$. Following Kennerberg and Wit [2025], for $A \in \mathscr{V}$, define the joint shift covariance function matrix,

$$K_A(s,t) = \begin{bmatrix} K_{A(1),A(1)}(s,t) & K_{A(1),A(2)}(s,t) & \dots & K_{A(1),A(p+1)}(s,t) \\ \vdots & \ddots & & \vdots \\ K_{A(p+1),A(1)}(s,t) & K_{A(p+1),A(2)}(s,t) & \dots & K_{A(p+1),A(p+1)}(s,t) \end{bmatrix}$$

where $K_{A(i),A(j)}(s,t) = \mathbb{E}[A_s(i)A_t(j)]$.

The shift set is defined as:

$$C^{\gamma}_{\mathscr{A}}(A) = \left\{ A' \in \mathscr{A} : \gamma K_A - K_{A'} \text{ is positive semi-definite} \right\} \tag{12}$$

### 3.1 Framework Comparison: Linear vs. FAR Operator

To understand the theoretical "roadblock" of extending functional worst-risk minimization to non-linear systems, it is highly interpretive to compare the original linear framework side-by-side with the proposed FAR architecture.

The Past: The Linear Framework (Kennerberg and Wit [2025]) In the original framework, the functional predictor is strictly linear. Consequently, the prediction risk under an intervention $A'$ can be evaluated exactly.

- **Predictor:** $\hat{Y}_{Lin} = \int \beta(t,\tau) X(\tau) d\tau$
- **Risk Function:** $R_{A'}(\beta) = \mathbb{E}\left[ ||Y^{A'} - \hat{Y}^{A'}_{Lin}||^2 \right]$

- **Worst-Risk Solution:** Because the operator is linear, the supremum of the risk over the shift set $C^{\gamma}_{\mathscr{A}}(A)$ resolves into an exact affine decomposition:

$$\sup_{A' \in C^{\gamma}_{\mathscr{A}}(A)} R_{A'}(\beta) = \frac{1}{2} R_{+}(\beta) + \left(\gamma - \frac{1}{2}\right) R_{\Delta}(\beta) \tag{13}$$

This exact decomposition allows the optimal projection kernel $\beta$ to be found globally via the inversion of empirical covariance matrices.

The New: Using the FAR Framework (Fan et al. [2015]) To model complex dynamic systems, we replace the linear predictor with the Functional Additive Regression operator, "compressing" the functional curves before applying a non-linear link function $g_j$.

- **Predictor:** $\hat{Y}_{FAR} = \sum_{j=1}^{p} g_j \left( \int \beta_j(t,\tau) X_j(\tau) d\tau \right)$
- **Risk Function:** $R^{FAR}_{A'}(g,\beta) = \mathbb{E}\left[ ||Y^{A'} - \hat{Y}^{A'}_{FAR}||^2 \right]$
- **Worst-Risk Solution:** Because the shift $A'$ is now mathematically nested inside the non-linear function $g_j$, the distributive property fails. The risk no longer decomposes algebraically:

$$\sup_{A' \in C^{\gamma}_{\mathscr{A}}(A)} R^{FAR}_{A'}(g,\beta) \neq \frac{1}{2} R_{+}(g,\beta) + \left(\gamma - \frac{1}{2}\right) R_{\Delta}(g,\beta) \tag{14}$$

Because the exact closed-form decomposition is invalidated, minimizing the worst-case risk for the FAR operator requires a fundamentally new approach. We need to establish a theoretical upper bound (proving that the error would not go to infinity) and follow a non-convex empirical optimization process.

## 3.2 Worst-Case Risk Upper Bound

Let $\mathscr{M}$ be our non-linear prediction operator under the FAR architecture, yielding predictions $\tilde{Y}^{A'} = \mathscr{M}(X^{A'})$. The prediction risk under shift $A'$ is $R_{A'}(\mathscr{M}) = \mathbb{E}[||Y^{A'} - \mathscr{M}(X^{A'})||^2_{L^2}]$.

To bound this risk, we analyze the residual error $E_{A'} = Y^{A'} - \mathscr{M}(X^{A'})$. Using the Fréchet derivative, we perform a first-order Taylor expansion of the error around the unshifted baseline state. Because the shift $A'$ and the baseline noise $\epsilon$ are independent, cross-terms vanish in expectation.

$$E_{A'} = \Phi(0) + \mathscr{D}[A'] + \mathscr{R}(A') \tag{15}$$

In this mathematical expansion:

- The **Residual Error ($E_{A'}$)** is the remaining, unexplained prediction error after the model $\mathscr{M}$ applies to the shifted data.
- The **Fréchet Derivative ($\mathscr{D}$)** is the generalization of the traditional derivative applied to infinite-dimensional functional spaces (Hilbert spaces). It measures how the non-linear operator reacts linearly to a tiny functional change.
- The **Taylor Remainder ($\mathscr{R}(A')$)** is the leftover, non-linear error that is not captured by the first-order Fréchet derivative.

This yields the Non-Linear Worst-Risk Upper Bound:

$$\sup_{A' \in C^{\gamma}_{\mathscr{A}}(A)} R_{A'}(\mathscr{M}) \leq R_O(\mathscr{M}) + \gamma \cdot \Omega(K_A, \nabla \mathscr{M}) + \mathscr{O}(L^2_{lip}) \tag{16}$$

where $R_O(\mathscr{M})$ is the baseline observational risk, $\Omega$ represents the specific sensitivity penalty of the predictor evaluated against the observed shift covariance $K_A$, and $L_{lip}$ is the Lipschitz constant of the non-linear link function $g$.

## 3.3 Proof of the Non-Linear Worst-Risk Upper Bound

We define the functional residual mapping $\Phi : L^2([T_1,T_2])^{p+1} \to L^2([T_1,T_2])$ that takes an external shift $A'$ and outputs the residual prediction error:

$$E_{A'} = \Phi(A') = Y^{A'} - \mathscr{M}(X + A') \tag{17}$$

Step 1: Fréchet Expansion of the Residual Error Assuming the non-linear link function $g$ estimated via smooth splines is Fréchet differentiable, we perform a first-order Taylor expansion of the residual error around the unshifted baseline state ($A' = 0$):

$$E_{A'} = \Phi(0) + \mathscr{D}[A'] + \mathscr{R}(A') \tag{18}$$

where $E_0 = \Phi(0)$ is the baseline prediction error driven strictly by the system noise $\epsilon_0$, $\mathscr{D} = \Phi'(0)$ is the linear, bounded Fréchet derivative operator evaluated at the baseline (dependent on $g'$ and $\beta$), and $\mathscr{R}(A')$ is the higher-order non-linear Taylor remainder.

Step 2: Expected $L^2$ Risk and Independence Substituting the expansion into the squared $L^2$ norm yields:

$$||E_{A'}||_{L^2}^2 = ||E_0||^2 + ||\mathscr{D}[A']||^2 + ||\mathscr{R}(A')||^2 + 2\langle E_0, \mathscr{D}[A']\rangle + 2\langle E_0 + \mathscr{D}[A'], \mathscr{R}(A')\rangle \tag{19}$$

Taking the expectation $\mathbb{E}[\cdot]$, and assuming the external shift has mean zero ($\mathbb{E}[A'] = 0$) and is independent of the internal system noise $\epsilon$, the expectation of the first cross-term vanishes:

$$\mathbb{E}\big[2\langle E_0, \mathscr{D}[A']\rangle\big] = 2\big\langle \mathbb{E}[E_0], \mathscr{D}\big[\mathbb{E}[A']\big]\big\rangle = 0 \tag{20}$$

The total expected risk simplifies to:

$$\mathbb{E}\big[||E_{A'}||^2\big] = R_O(\mathscr{M}) + \mathbb{E}\big[||\mathscr{D}[A']||^2\big] + \mathbb{E}\big[||\mathscr{R}(A')||^2 + 2\langle E_0 + \mathscr{D}[A'], \mathscr{R}(A')\rangle\big] \tag{21}$$

Step 3: Bounding the First-Order Variance We isolate the first-order variance term injected by the shift $A'$. Using the properties of the adjoint operator $\mathscr{D}^*$ and the cyclic property of the trace with the covariance operator $K_{A'} = \mathbb{E}[A' \otimes A']$:

$$\mathbb{E}\big[||\mathscr{D}[A']||^2\big] = \mathrm{Tr}(\mathscr{D}^*\mathscr{D}K_{A'}) = \mathrm{Tr}(\mathscr{D}K_{A'}\mathscr{D}^*) \tag{22}$$

By the definition of our shift set, $K_{A'} \preceq \gamma K_A$. Because $\mathscr{D}^*\mathscr{D}$ is a positive operator, it preserves this inequality:

$$\mathrm{Tr}(\mathscr{D}K_{A'}\mathscr{D}^*) \leq \mathrm{Tr}(\mathscr{D}(\gamma K_A)\mathscr{D}^*) = \gamma \mathrm{Tr}(\mathscr{D}K_A\mathscr{D}^*) \tag{23}$$

We define this upper bound as $\Omega(K_A, \nabla\mathscr{M}) := \mathrm{Tr}(\mathscr{D}K_A\mathscr{D}^*)$, yielding:

$$\mathbb{E}\big[||\mathscr{D}[A']||^2\big] \leq \gamma\Omega(K_A, \nabla\mathscr{M}) \tag{24}$$

Step 4: Bounding the Non-Linear Remainder To ensure the error does not explode arbitrarily, we bound the Taylor remainder $\mathscr{R}(A')$. In the FAR framework, the non-linearity is constrained to the 1-dimensional link function $g$. Assuming $g$ has a bounded second derivative (or is uniformly Lipschitz continuous with constant $L_{lip}$), the remainder is strictly proportional to the squared magnitude of the shift projection:

$$||\mathscr{R}(A')||_{L^2}^2 \leq \frac{1}{4}L_{lip}^2||\langle\beta, A'\rangle||^4 \tag{25}$$

Because the projection kernel $\beta$ is bounded and normalized ($||\beta||_2 = 1$), the remainder is directly bounded by the trace of the shift covariance. Applying the environment budget ($K_{A'} \preceq \gamma K_A$), the entire expectation of the remainder mathematically collapses into a strictly bounded constant $\mathscr{O}_{rem}$:

$$\mathbb{E}\big[||\mathscr{R}(A')||^2 + 2\langle E_0 + \mathscr{D}[A'], \mathscr{R}(A')\rangle\big] \leq \mathscr{O}_{rem}(L_{lip}, \gamma K_A) \tag{26}$$

Step 5: Conclusion and Assembly Substituting the first-order covariance bound (Equation 24) and the non-linear remainder bound (Equation 26) into the expected risk, we arrive at the guaranteed upper limit for any single shift $A'$. Taking the supremum over all allowable shifts $A' \in C_{\mathscr{A}}^{\gamma}(A)$ establishes the final Non-Linear Worst-Risk Upper Bound:

$$\sup_{A' \in C_{\mathscr{A}}^{\gamma}(A)} R_{A'}(\mathscr{M}) \leq R_O(\mathscr{M}) + \gamma \cdot \Omega(K_A, \nabla\mathscr{M}) + \mathscr{O}_{rem}(L_{lip}, \gamma K_A) \tag{27}$$

This final bound relies on several key theoretical constraints functioning together:

- The **Lipschitz Constant ($L_{lip}$)** acts as a strict mathematical bound on how fast the non-linear function $g$ can change. If the function's slope is restricted, the Taylor remainder cannot explode to infinity.
- The **Supremum (**sup**)** is the least upper bound of a set, representing the absolute "worst-case" scenario among all allowable future shifts within the budget.
- The **Positive Semi-Definite Trace ($\Omega$)** is the trace of the resulting covariance matrices. When a matrix is positive semi-definite, its trace provides a guaranteed upper bound on the variance inflation caused by the shift.

# 4 The Empirical Objective

Because the theoretical population risk cannot be linearly decomposed under the $\mathscr{T}_{NL}$ operator, we must reformulate the worst-risk minimization as an empirical, joint optimization problem. However, before the algorithm can navigate the loss landscape, we must resolve a fundamental mathematical ambiguity "built-in" to the FAR architecture.

The FAR operator predicts the response via a projection index $u = \langle \beta, X \rangle$, which is fed into a non-linear link function $g(\cdot)$:

$$\hat{Y} = g(\langle \beta, X \rangle) \tag{28}$$

Without constraints, this operator is structurally unidentifiable due to the "scaling drift". For any non-zero scalar constant $c$, we can define a new projection kernel $\tilde{\beta} = c\beta$ and a correspondingly scaled non-linear function $\tilde{g}(u) = g(u/c)$. Substituting these yields the exact same prediction:

$$\tilde{g}(\langle \tilde{\beta}, X \rangle) = g\left( \frac{\langle c\beta, X \rangle}{c} \right) = g(\langle \beta, X \rangle) = \hat{Y} \tag{29}$$

This proves that an infinite number of $(\tilde{g}, \tilde{\beta})$ pairs can produce the identical output. If optimized without constraints, the algorithm will infinitely shrink the amplitude of the non-linear function $g$ while simultaneously making the projection kernel $\beta$ go to infinity, at the end crashing the estimator.

## 4.1 Enforcing Identifiability

To eliminate this infinite scaling ambiguity and force mathematical identifiability, we adopt the strict constraint proposed by Fan et al. [2015], requiring the $L_2$ norm of the projection kernel to equal 1:

$$||\beta||_2 = 1 \tag{30}$$

This restricts the infinite space of potential scalars to just $c = 1$ or $c = -1$. To resolve the remaining sign ambiguity, we apply a positive trace convention. If the sum of the diagonal elements of the projection matrix is negative, the entire kernel is multiplied by $-1$.

While the $||\beta||_2 = 1$ condition fixes the internal scaling of the single-index projection, the additive nature of the $\mathscr{T}_{NL}$ operator (which sums multiple non-linear functions $\sum_{j=1}^{p} g_j$) introduces a second layer of unidentifiability. Without further constraints, the model suffers from location drift, where arbitrary additive constants can be passed between the functions $g_j$ without changing the final prediction $\hat{Y}$. To ensure the functions $g_j$ are uniquely identified, we must enforce a strict centering constraint:

$$\mathbb{E}[g_j(u)] = 0 \quad \text{for all } j = 1, \ldots, p \tag{31}$$

By forcing every non-linear function to have a mean of zero, any global baseline offset in the functional system is then absorbed by a single, global intercept term. This theoretical requirement perfectly explains the empirical use of centered B-splines in the later estimation steps.

## 4.2 The Basis Expansion

To implement the empirical estimation of the unknown non-linear link functions $g_j$, we utilize a B-spline basis expansion. The non-linear transformation for the projected index $u = \langle \beta, X_j \rangle$ is mathematically approximated as a linear combination of basis functions:

$$g_j(u) \approx \sum_{k=1}^{K} \theta_{jk} B_k(u) \tag{32}$$

where $B_k(u)$ are the B-spline basis functions with $K = 5$ degrees of freedom, and $\theta_{jk}$ are the corresponding unknown coefficients. For a single functional covariate $X_j$, we define the coefficient vector as $\boldsymbol{\theta}_j = (\theta_{j1}, \ldots, \theta_{jK})^T$. By substituting this basis expansion into the empirical Mean Squared Error, the problem transitions from an intractable infinite-dimensional functional optimization over $g_j$ to a manageable finite-dimensional vector and matrix optimization problem over the coefficient vectors $\boldsymbol{\theta}_j$.

## 4.3 The Empirical Worst-Risk Formulation

With the parameters strictly identifiable, we define the empirical worst-risk objective function. Using the hyperparameter $\gamma \in [0, 1]$ to manage the trade-off between the observed environments, the algorithm minimizes the total loss directly over the observed (Obs) and interventional (Int) data points:

$$\mathscr{L}(\beta, g) = (1 - \gamma) \cdot MSE_{Obs}(\beta, g) + \gamma \cdot MSE_{Int}(\beta, g) \tag{33}$$

where the mean squared errors are defined as:

$$MSE_{Obs}(\beta, g) = \frac{1}{n_{obs}} \sum_{i=1}^{n_{obs}} \left(Y_i^{obs} - g(\langle \beta, X_i^{obs} \rangle)\right)^2 \tag{34}$$

$$MSE_{Int}(\beta, g) = \frac{1}{n_{int}} \sum_{i=1}^{n_{int}} \left(Y_i^{int} - g(\langle \beta, X_i^{int} \rangle)\right)^2 \tag{35}$$

This adapts the penalized least squares optimization approach used by Fan et al. [2015] to account for our interventional risk penalty.

Minimizing this joint loss objective guarantees that the resulting parameters will respect the expected risk upper bound established in Section 3.2.

# 5 Empirical estimation

In a linear system, the worst-risk minimizer can be solved globally via a single matrix inversion. The transition to the FAR framework fundamentally destroys this convexity. Because the unknown projection kernel $\beta$ is nested inside the unknown non-linear link function $g$, they depend on each other and cannot be solved simultaneously.

## 5.1 The Optimization Process

In the process of developing the algorithm, different approaches attempted to optimize the directional projection kernels and the complex B-spline shapes simultaneously in a single large loop. However, this often led to instability, overfitting of local noise, and extrapolation problems at the data boundaries. To safely traverse the non-convex loss landscape, we resolve these issues by splitting the algorithm into a decoupled multi-phase process: employing an Alternating Coordinate Descent purely for directional estimation, separated from the penalized non-linear shape fitting.

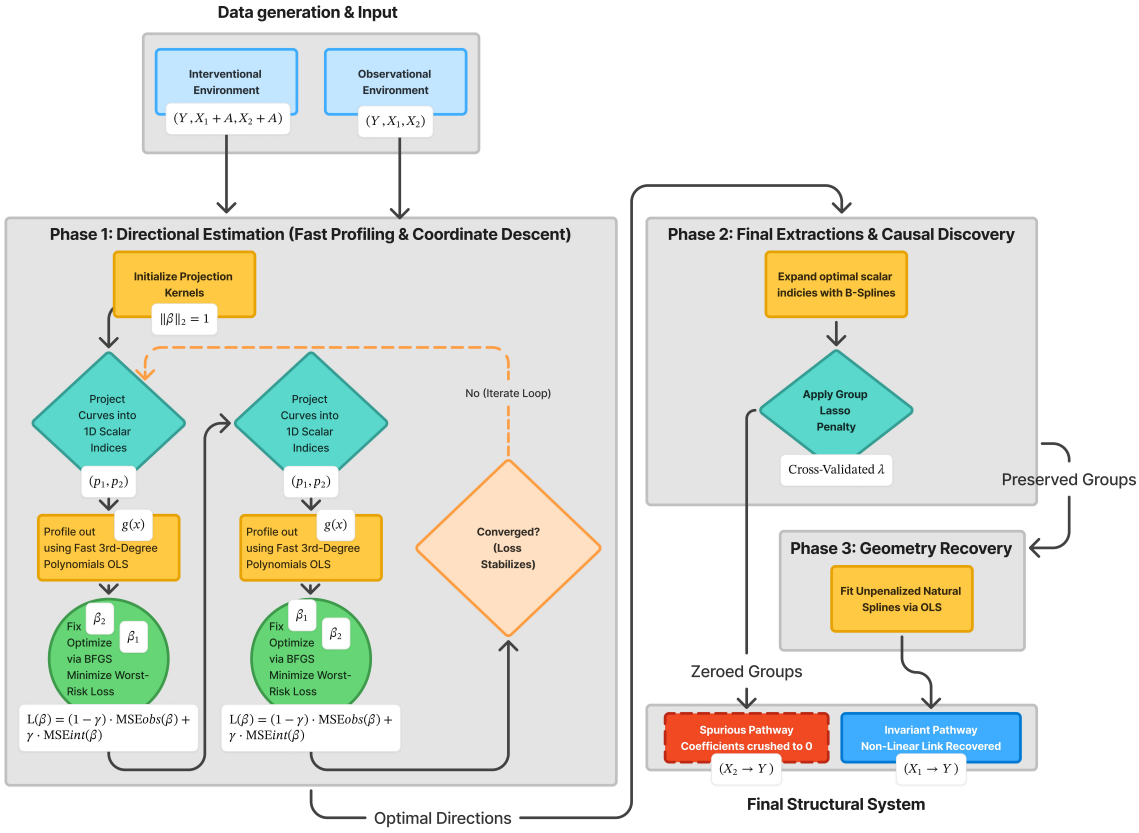



**Figure 1.** The diagram of the algorithm's flow.

#### 5.1.1 Phase 1: Directional Estimation (Fast Polynomial Profiling)

In the first phase, we identify the optimal projection kernel $\beta$ without attempting to fit the complex spline shape. Because the BFGS optimization algorithm requires calculating the loss surface and its gradients hundreds of times per second, fitting full B-splines inside the gradient loop is computationally explosive and highly unstable. Instead, the algorithm employs a fast polynomial profiling technique:

- **Initialization:** The projection kernels are initialized randomly and normalized to satisfy $\|\beta\|_2 = 1$. The data is pooled and weighted by the $\gamma$ worst-risk parameters.
- **Alternating Projection:** Instead of optimizing all parameters at once, the algorithm employs block coordinate descent. This builds upon the highly efficient coordinate descent algorithm originally established for optimizing the FAR framework by Fan et al. [2015]. It holds the projection matrix $C_2$ fixed to isolate and optimize the rotation of $C_1$, and then alternates, fixing $C_1$ while optimizing $C_2$. The data is projected into temporary 1-dimensional scalar indices $p_1$ and $p_2$.
- **Fast Profiling:** Instead of full splines, we construct a compact, fast design matrix using 3rd-degree polynomials evaluated on both indices: $X_{design} = [1, p_1, p_2, p_1^2, p_2^2, p_1^3, p_2^3]$.
- **Optimization (BFGS):** Within each alternating step, the non-linear link function $g(\cdot)$ is instantaneously "profiled out" using a highly optimized, pooled Ordinary Least Squares (OLS) solver (`lm.fit`) on the joint datasets. This delivers a relatively fast, smooth loss gradient that allows the quasi-Newton optimizer to efficiently traverse the non-linear space and lock onto the true causal directions with a series of iterations.

#### 5.1.2 Phase 2: Final Extraction via Group Lasso

The critical architectural decision in non-linear causal discovery is evaluating causal edges jointly rather than in isolation. If an algorithm evaluates a spurious correlation in isolation, the mathematical drive to minimize prediction error will overpower the interventional worst-risk penalty.

Once the optimal projection kernels converge in Phase 1, the algorithm permanently fixes the directions and extracts the 1-dimensional scalar indices, $p_1$ and $p_2$. These indices are expanded using a B-spline basis with 5 degrees of freedom. The finalized design matrix is fed into a cross-validated Group Lasso model [Huang et al., 2012].

Formally, the penalized objective function used in Phase 2 combines the joint worst-risk empirical loss $\mathscr{L}(\beta, g)$ with the Group Lasso penalty:

$$\min_{\boldsymbol{\theta}} \left\{ \mathscr{L}(\beta, g) + \lambda \sum_{j=1}^{p} \|\boldsymbol{\theta}_j\|_2 \right\} \tag{36}$$

This formulation provides a theoretical justification connecting it directly to causal discovery. Because the entire non-linear effect of the variable $X_j$ is represented by the group of basis coefficients $\boldsymbol{\theta}_j$, the $\ell_2$-norm penalty evaluates the causal edge jointly. Instead of penalizing individual wiggles in the spline, this structurally forces the entire non-linear pathway of a spurious, unstable variable go to zero.

Because the true invariant cause perfectly predicts $Y$ without having an interventional penalty, the spurious variable becomes mathematically redundant. The Group Lasso executes non-linear causal discovery by zeroing out the basis groups associated with unstable, environment-dependent variables.

#### 5.1.3 Phase 3: Non-Linear Link Function Recovery

With the causal architecture identified and the spurious pathways penalized to zero, the final step isolates the structural geometry of the true causal links. The algorithm calculates the matching projected index based on the final signs and normalized scale. To yield a clean, unpenalized visualization of the recovered causal mechanism, the algorithm fits an unpenalized Natural Spline (NS) using Ordinary Least Squares over the valid projection index. This helps us in decoupling the penalized causal search from the final shape extraction, preventing boundary extrapolation errors.

### 5.2 Hyperparameter Tuning

The success of this empirical estimation relies on the interplay between two critical tuning parameters in this framework: $\gamma$ and $\lambda$.

- $\gamma \in [0, 1]$: The causal regularization/environmental parameter. It controls the worst-risk robustness, telling the algorithm how aggressively it should penalize the interventional variance relative to baseline observational fit.

- $\lambda > 0$: The Group Lasso sparsity penalty parameter. It controls the complexity and selection of the non-linear bases, ensuring that unstable causal pathways are eliminated.

To navigate the computational trade-offs, these parameters are tuned sequentially rather than through a computationally explosive 2D grid search. The environmental parameter $\gamma$ is evaluated over a targeted grid (e.g., $0.5, 0.8, 0.9$) or selected based on the desired level of robustness and stability. For each formulated $\gamma$, the Group Lasso penalty $\lambda$ is independently and dynamically chosen via cross-validation directly within the Group Lasso fitting procedure. This sequential strategy maintains a computationally tractable optimization path while ensuring strict sparsity enforcement.

# 6 Simulation and Causal Discovery

To test the mathematical framework and the algorithm, we simulated the DAG system under two distinct non-linear systems: a complex Sinusoidal topology and a Cubic polynomial topology. This helps us to validate the coordinate descent algorithm's capability to isolate true causes under different non-proportional relationships.

## 6.1 Data Generation Process

We construct a synthetic functional system conforming to a strict Directed Acyclic Graph (DAG) specified by the causal relationships $X_1 \to Y \to X_2$. Here, $X_1$ serves as the true invariant functional cause of the target process $Y$, whereas $X_2$ represents a spurious variable whose association with $Y$ is purely confounded by the reverse causal relation.

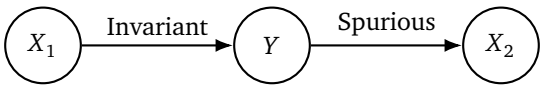



**Figure 2.** Directed Acyclic Graph (DAG) of the simulated functional structural equation model. $X_1$ acts as the true invariant cause, while $X_2$ is a consequence.

The functional curves are generated using an orthonormal basis expansion with $d = 5$ dimensions over a compact time domain $t \in [0, 1]$. The system is evaluated under two distinct functional setups to evaluate the estimator's capability of detecting different non-proportional shapes.

Sinusoidal Relation: The relationships are governed by trigonometric non-linear link functions:

$$g_{x_1 \to y}(x) = x + \sin(2x) \tag{37}$$
$$g_{y \to x_2}(x) = -x + \sin(x) \tag{38}$$

Cubic Relation: The relationships are governed by polynomial non-linear link functions:

$$g_{x_1 \to y}(x) = x + 0.1x^3 \tag{39}$$
$$g_{y \to x_2}(x) = -x + 0.1x^3 \tag{40}$$

The data-generating process simulates two distinct environments:

- **Observational Environment:** Functional curves are generated strictly based on internal exogenous system noise $\epsilon \sim \mathcal{N}(0, 1)$, preserving the baseline equilibrium of the system.

- **Interventional Environment:** An external distributional shift $A$ is additively injected into the exogenous variables. The shift is modeled as a random perturbation drawn from $A \sim \mathcal{N}(\mu_A, \sigma_A^2)$ with $\mu_A = 3.0$ and $\sigma_A = 1.0$. Crucially, this intervention is applied directly to $X_1$ and $X_2$, while the structural relation determining $Y$ remains completely invariant across environments.

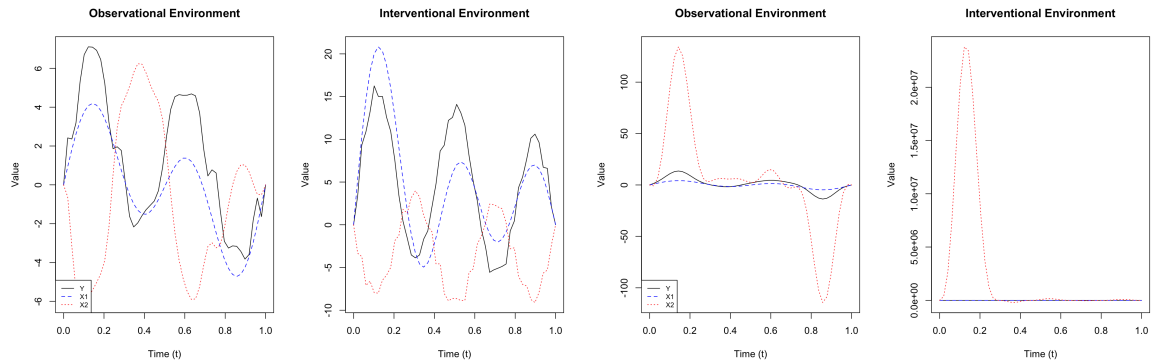



**Figure 3.** Observational versus Interventional environments. **Left:** Sinusoidal. **Right:** Cubic. These plots demonstrate the magnitude of the distributional shifts injected into the system. Note how the injected external shift alters the variance and amplitude of $X_1$ and $X_2$. While both $X_1$ and the spurious $X_2$ may appear highly predictive of the target $Y$ within a single environment, only the invariant cause $X_1$ maintains a stable structural relationship across both regimes.

The estimation algorithm is then run on these generated datasets, with the interventional risk penalty parameter set to $\gamma = 0.8$.

## 6.2 Recovered Projection Kernels

The most straightforward representation of functional causal discovery lies in examining the recovered projection kernels $\beta(t, \tau)$ and their corresponding 2D heatmaps.

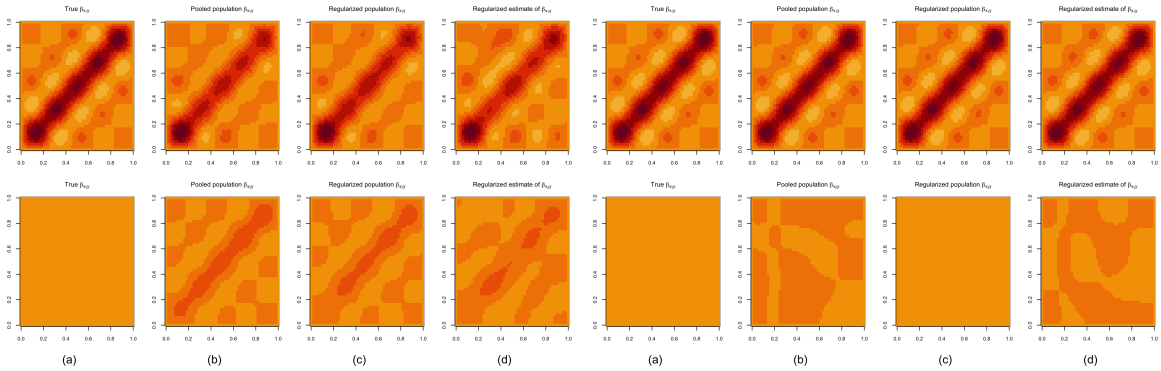



**Figure 4.** Pairwise-normalized estimated heatmaps. **Left:** Sinusoidal system. **Right:** Cubic system. In both environments, the regularized estimation algorithms aim to structurally suppress the spurious variable $X_2$ (bottom row) toward zero, preserving the true causal kernel of $X_1$ (top row).
The Pooled population heatmap means setting the gamma to 0.5, thus not considering the interventional environment tradeoff, and doing the basic pooled risk minimization. "Regularized" plots on the other hand show us the results with gamma set to 0.9, providing the difference in estimation when considering the shift more. "Population" results are the estimation done on the larger sample size (400), whilst the "estimate" plots are the results of the smaller sample (100) optimization.

Because the algorithm actively penalizes the interventional risk $\gamma$ injected into $X_2$ via the external shift $A$, the coordinate descent rotates the weights towards $X_1$, heavily suppressing the "false positive" mapping. However, as seen in the pairwise-normalized heatmaps (Figure 4), the regularized estimation does not yield a mathematically perfect zero-matrix. Minor residual artifacts remain, highlighting the difficulty of achieving absolute sparsity in a non-convex functional space. To have a more native view on the estimation result, we can visualize and observe the full 3D surfaces of these kernels.




**Figure 5. Sinusoidal System (Invariant Cause $X_1 \to Y$):** True kernel (left) vs Estimated kernel (right). The algorithm tries to profile the underlying relation, capturing the complex dependence.




**Figure 6. Cubic System (Invariant Cause $X_1 \to Y$):** True kernel (left) vs Estimated kernel (right). The estimator successfully recovers the invariant structural shape.

Figures 5 and 6 show the estimation of the true causal pathway. The algorithm's BFGS inner loop traverses the non-convex loss space to find the optimal rotation, yielding estimated surfaces that mathematically resemble the true structural kernels.

While the results are not perfect, especially with the sinusoidal relation estimation, the algorithm closely captures the underlying landscape, despite the challenge of solving a highly complex non-convex optimization problem.




**Figure 7. Sinusoidal System (Spurious Correlate $X_2 \to Y$):** While a naturally occurring spurious correlation landscape exists due to the downstream DAG relation, the estimation algorithm actively penalizes it, shrinking the estimated impact toward zero.




**Figure 8. Cubic System (Spurious Correlate $X_2 \to Y$):** True correlation (left) vs Estimated kernel (right). The algorithmic penalty heavily suppresses the spurious weights, effectively neutralizing the variable for causal discovery.

Figures 7 and 8 serve as the visual proof of functional causal discovery, while also illustrating the practical limits of empirical estimation. In a standard observational regression, the model would attempt to map the spurious correlation in order to minimize the loss. However, driven by the interventional risk penalty $\gamma$, the functional worst-risk minimizer recognizes $X_2$ as unstable.

The Group Lasso extraction successfully flattens the macro-structure of the estimated kernel, functionally cutting the spurious edge from the predicted system. Yet, closer inspection of Figure 8 reveals minor residual waviness across the estimated surface. This is an "attribute" of non-convex optimization over complex basis expansions; while the penalty shrinks the group coefficients toward zero, navigating out of deep local minima to achieve a perfectly flat Euclidean plane across the entire 2D domain is computationally challenging. Nonetheless, the penalty is severe enough to neutralize the variable's predictive power during out-of-sample shifts.

## 6.3 Recovery of Non-Linear Link Functions

We can observe the algorithm's capacity to estimate the geometry of $g(x)$ by plotting the valid projected index against the observed target variable. As demonstrated in Figure 9, the estimator successfully tracks the core structure of both the sinusoidal and cubic generative mechanisms within the high-density interior regions of the data. By utilizing robust unpenalized natural splines in the final extraction phase, the algorithm maps the functional coordinate projections without hallucinating structural pathways from the random noise $\epsilon$.

However, these plots also reveal the inherent limitations of spline-based estimation at the domain boundaries. In Figure 9, the estimated spline (red) noticeably diverges from the true invariant mechanism (blue dashed line) at the extreme tails of the projected index. In the sinusoidal system (left), the spline smooths over the final trigonometric dip, while in the cubic system (right), it slightly underestimates the steepness of the polynomial trajectory at the edges. Because natural splines are constrained to be linear beyond the boundary knots to prevent catastrophic divergence, they naturally lack the flexibility (and the local data density) required to perfectly map extreme non-proportional relation at the absolute margins of the environment.

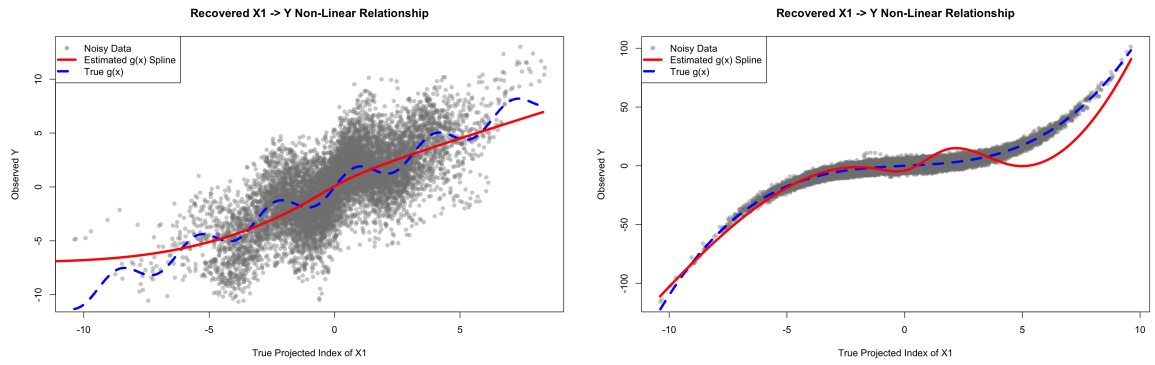



**Figure 9.** Recovery of the non-linear generative mechanism $g_{x_1 \to y}$. **Left:** The true sinusoidal link's direction and the form successfully tracked by the Ridge-penalized B-spline (red). **Right:** The cubic relationship recovered. The model attempts to recover the non-proportional topology despite massive variance injections from the functional indexes.

## 6.4 Interventional Predictions and Error Analysis

To thoroughly document the algorithmic performance out-of-sample, we evaluate how the models handle severe environmental interventions. While the structural geometry is largely recovered, the transition to non-linear functional operators introduces specific predictive limitations that are absent in strictly linear frameworks.

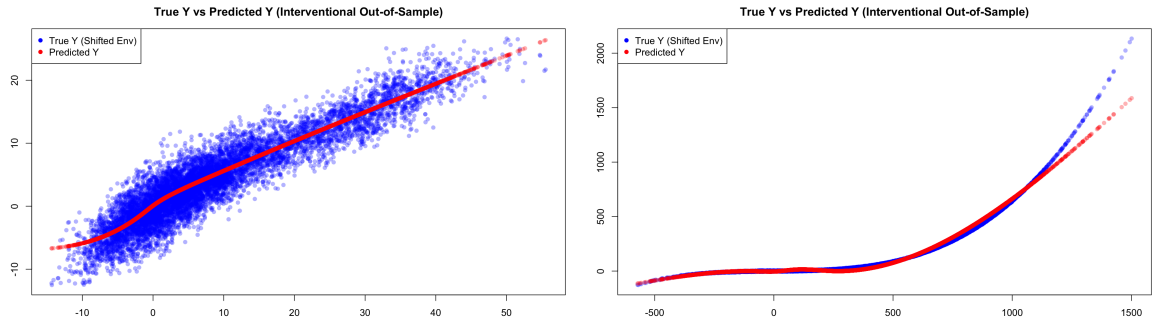



**Figure 10.** Prediction tracking over time in the interventional setting. **Left:** Sinusoidal. **Right:** Cubic. The algorithm predicts the target variable (red line) accurately despite the severe shifts in the underlying predictors, tracking the true interventional observations (blue) closely within the interior of the functional interval.

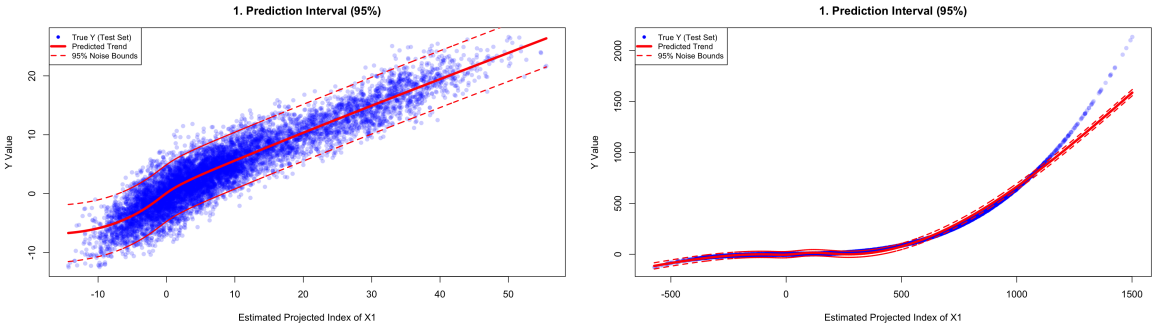



**Figure 11.** Displaying the error bounds. **Left:** Sinusoidal. **Right:** Cubic. This plot demonstrates how the predicted trend tracks the true data points in the interior, while highlighting the increased prediction variance at the domain boundaries.

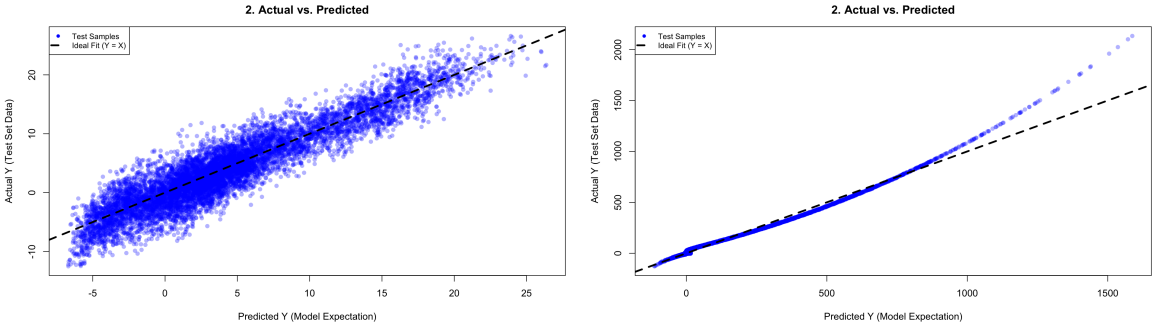



**Figure 12.** Actual vs. Predicted scatter plots. **Left:** Sinusoidal. **Right:** Cubic. Points mapping closely to the $y = x$ diagonal demonstrate the accuracy of the predictions against the actual values in the test set.

As observed in the prediction tracking (Figure 10) and the actual versus predicted scatter plots (Figure 12), the algorithm successfully captures the primary target variable trend despite severe shifts in the underlying predictors. By isolating the invariant causal path ($X_1 \to Y$) and applying the Group Lasso penalty to discard the spurious path ($X_2 \to Y$), the model effectively ignores the variance inflation caused by the intervention $A$. In the high-density interior regions of the data, the predictions map closely to the ideal $y = x$ diagonal.

However, predictive performance naturally degrades near the domain boundaries. Because the non-convex optimization relies on a finite B-spline basis to map the non-linear links, the model has inherent extrapolation instability when the severe interventional shift pushes the projected indices into the outlying, previously unobserved coordinate spaces.

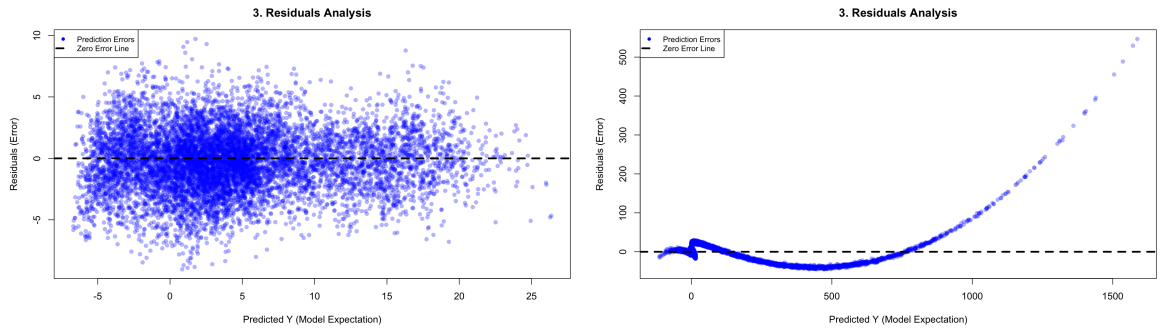



**Figure 13.** Residual Analysis. **Left:** Sinusoidal. **Right:** Cubic. While errors remain centered around zero in the high-density interior, the cubic residual plot reveals expected boundary variance inflation under severe shifts.

This boundary degradation is clearly visible in the residual analysis (Figure 13). While a linear worst-risk minimizer might yield uniformly distributed homogeneous errors, the non-linear estimator exhibits distinct heterogeneous errors under severe shifts. Specifically, in the cubic system's residual analysis, while the errors are cleanly centered around zero in the interior, the residual variance visibly inflates at the upper and lower bounds.

This U-shaped residual expansion is a direct empirical manifestation of the higher-order Taylor remainder $\mathfrak{R}(A')$ established in Section 3.2. As the shift magnitude increases at the edges of the spline's knot sequence, the unpenalized natural spline struggles to strictly contain the steep gradients of the underlying polynomial function, leading to boundary variance.

Ultimately, these diagnostic plots demonstrate a successful, yet practically constrained, execution of functional causal discovery. The algorithm sacrifices some precision at the boundaries, being an expected computational trade-off when solving a highly non-convex objective with basis expansions, but still successfully prevents catastrophic extrapolation failure by zeroing out the spurious environmental areas.

# 7 Conclusion

In this thesis, we extended the functional worst-risk minimization framework to deal with the non-linear structural equation models. While the preceding work of Kennerberg and Wit [2025] established this foundation for purely linear relationships, we integrated the Functional Additive Regression (FAR) operator from Fan et al. [2015] to model complex, real-world like dynamics.

As we can observe, transitioning to a non-linear system introduces the fundamental "roadblock" of the distributive failure. Because the interventional shift is mathematically nested inside the non-linear link function, an exact worst-risk decomposition is no longer possible. Our key theoretical contribution was overcoming this barrier by establishing a mathematical upper bound. This bound proves that even under strong and unseen interventional shifts, the worst-case prediction error is strictly contained and prevented from exploding to infinity.

To translate this theoretical bound into an empirical reality, we had to manage a highly non-convex optimization problem. We solved this by breaking the algorithm into three different phases. It begins with fast polynomial profiling to lock in the projection kernels, utilizes a cross-validated Group Lasso to drop false variables, and concludes by fitting an unpenalized natural spline to reveal the true non-linear function's geometry. Through simulations on both sinusoidal and cubic systems, we demonstrated that this approach successfully recovers the invariant causal path ($X_1 \to Y$) while actively trying to suppress the spurious correlates ($X_2 \to Y$). In respect that, as we mentioned, relying on a finite B-spline basis may introduce some extrapolation instability at the domain boundaries under extreme shifts, the algorithm successfully maintains robust predictive accuracy in the functional interior.

## 7.1 Future Work

While working on this thesis, we tried the approach on different operator definitions. Some of them, more theoretically sound, provided better identifiability context, introducing less constraints to the system, but, on the other

hand, bringing the severe dimensionality and complexity issues when considering the estimation process. Our chosen Single-Index model was a trade-off being able to capture the nonlinear dynamics, while overcoming the curse of dimensionality and allowing us to have a computationally tractable estimation technique.

Whilst this approach provided us with both the theoretical discovery and the empirical part insightful enough, using other more "complete" theory-wise operator definitions is definitely the main goal when talking about future extensions and scientific contribution.

To summarize, future research should focus on addressing the limitations of the single-index architecture, replacing it with a more theoretically sound and comprehensive functional operator. This transition would require overcoming significant mathematical and computational hurdles to balance theoretical completeness with the complexities of high-dimensional estimation, yielding deeper structural insights.

# References


Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. Invariant risk minimization, 2020. URL https://arxiv.org/abs/1907.02893.

Yingying Fan, Gareth M. James, and Peter Radchenko. Functional additive regression. *The Annals of Statistics*, 43(5), October 2015. ISSN 0090-5364. doi: 10.1214/15-aos1346. URL http://dx.doi.org/10.1214/15-AOS1346.

Jian Huang, Patrick Breheny, and Shuangge Ma. A selective review of group selection in high-dimensional models. *Statistical Science*, 27(4), November 2012. ISSN 0883-4237. doi: 10.1214/12-sts392. URL http://dx.doi.org/10.1214/12-STS392.

Gareth James and Bernard Silverman. Functional adaptive model estimation. *Journal of the American Statistical Association*, 100, 03 2004. doi: 10.1198/016214504000001556.

Lucas Kania and Ernst Wit. Causal regularization: On the trade-off between in-sample risk and out-of-sample risk guarantees, 2025. URL https://arxiv.org/abs/2205.01593.

Philip Kennerberg and Ernst C. Wit. Functional worst risk minimization, 2025. URL https://arxiv.org/abs/2412.00412.

Hans-Georg Müller and Fang Yao. Functional additive models. *Journal of the American Statistical Association*, 103: 1534 – 1544, 2008. URL https://api.semanticscholar.org/CorpusID:1927777.

Jonas Peters, Peter Bühlmann, and Nicolai Meinshausen. Causal inference using invariant prediction: identification and confidence intervals, 2015. URL https://arxiv.org/abs/1501.01332.

Dominik Rothenhäusler, Nicolai Meinshausen, Peter Bühlmann, and Jonas Peters. Anchor regression: heterogeneous data meets causality, 2020. URL https://arxiv.org/abs/1801.06229.